

COMP393 I/3932

Individual Project 25/26

Project organisation



Tentative topics/titles

Student	Pre	Topic
Alexander Del Brocco		
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Leyna Tiji		
Araav Bhardwaj		
Yusuf Demir		
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Project timeline & Next steps

Semester 1 (Project planning, background reading, identification of data and methods)

Week 2	Supervisor allocation
Week 6	Submission of Project Outline and Specification
Week 6	Assessor allocation
Week 9	Feedback on Project Outline and Specification
Every two weeks	Supervision meeting (supervisor-driven sessions + discussion)

Semester 2 (Implementation, experiments and testing, final report writing)

Week 4	Arrange meeting with Assessor
Week 7	Meeting with Assessor
Week 10	Submit Final Report
Every two weeks	Supervision meeting (student-driven short updates)

Project #1: Detecting and Correcting for Shortcut Learning in Medical Applications

Context: Machine learning is increasingly being used to train models for assisting diagnosis in medical applications. While powerful, ML models can suffer from bias or misleading predictions when applied outside of the training data (lack of generalisability). Shortcut learning refers to a phenomena where the model learns to make predictions based on spurious or misleading correlations in the data. A typical example is that age is often identified by ML models as a predictor for Alzheimer's disease, since advancing age strongly correlates with the probability of having AD. However, age has no causal relationship to AD. Another related phenomena is the so-called "machine learning horse" where the model observes irrelevant background features and uses them to make predictions, giving the appearance of high performance that will likely not generalise beyond the training data.

This project aims at: 1) setting up some standard ML problems in image-guided diagnosis that are known to be prone to shortcut learning as test cases, 2) performing a literature search on strategies of identifying and correcting for shortcut learning, and 3) developing and evaluating methods for correcting shortcut learning in the chosen test cases and demonstrating that better generalisability is achieved afterwards.

Background papers: Brown et al. 2023

Dataset: Various depending on test case, for example ADNI for Alzheimer's disease

Project #1: Detecting and Correcting for Shortcut Learning in Medical Applications

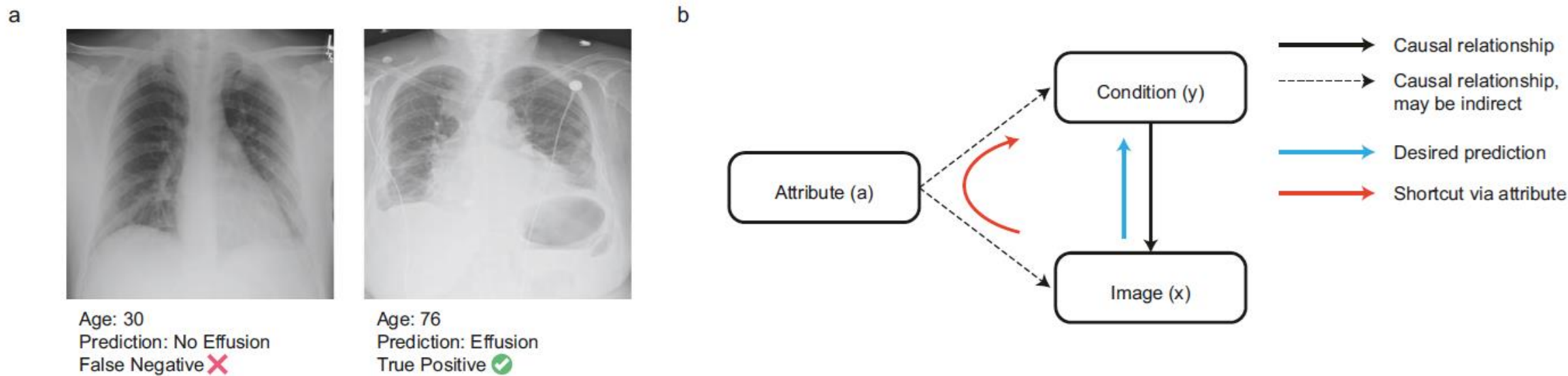


Fig. 1 | Is a model unfair due to shortcutting? **a** Examples of correct and incorrect predictions that may be influenced by age shortcut learning in a chest X-ray application detecting Effusion. In this example, the prediction is incorrect for a patient with atypical age, raising the possibility of shortcut learning. **b** Simplified diagram illustrating how shortcutting may occur. In this schematic, the presence or absence of a particular condition y will produce changes in the image x ; we

therefore wish to train a model that can predict y , given x (blue arrow). However, an attribute a , such as age, may alter both the risk of developing a given condition, as well as the image. This need not be a directly direct relationship (dotted arrows). A model may learn to predict the presence of a condition by using the attribute (red arrow). When these correlations are not considerably beneficial for model performance, we consider that this path is (at least partly) a shortcut.

Project #2: Image-quality Metrics for Machine Learning in Medical Images

Context: The goal of this project is to investigate the selection of image quality metrics for medical images in various machine learning applications. Image processing algorithms are commonly used to analyze medical images, and the quality of these images is crucial for accurate clinical diagnoses. The most popular image quality metrics are either basic pixel-wise metrics, like PSNR, which primarily measure noise, or perceptual quality metrics derived from studies on natural images, not specifically on medical images. Moreover, while generative AI is increasingly being applied to medical imaging, existing metrics for evaluating the quality of synthetic medical images may not be well-suited for this purpose.

This project could involve selecting a range of imaging datasets and corresponding deep learning tasks, conducting a comprehensive evaluation of current image quality metrics, and even proposing entirely new metrics tailored to medical images.

Background papers: Alaa et al 2022, Deo et al 2025

Dataset: To be selected by the student as part of the project specification

Project #2: Image-quality Metrics for Machine Learning in Medical Images

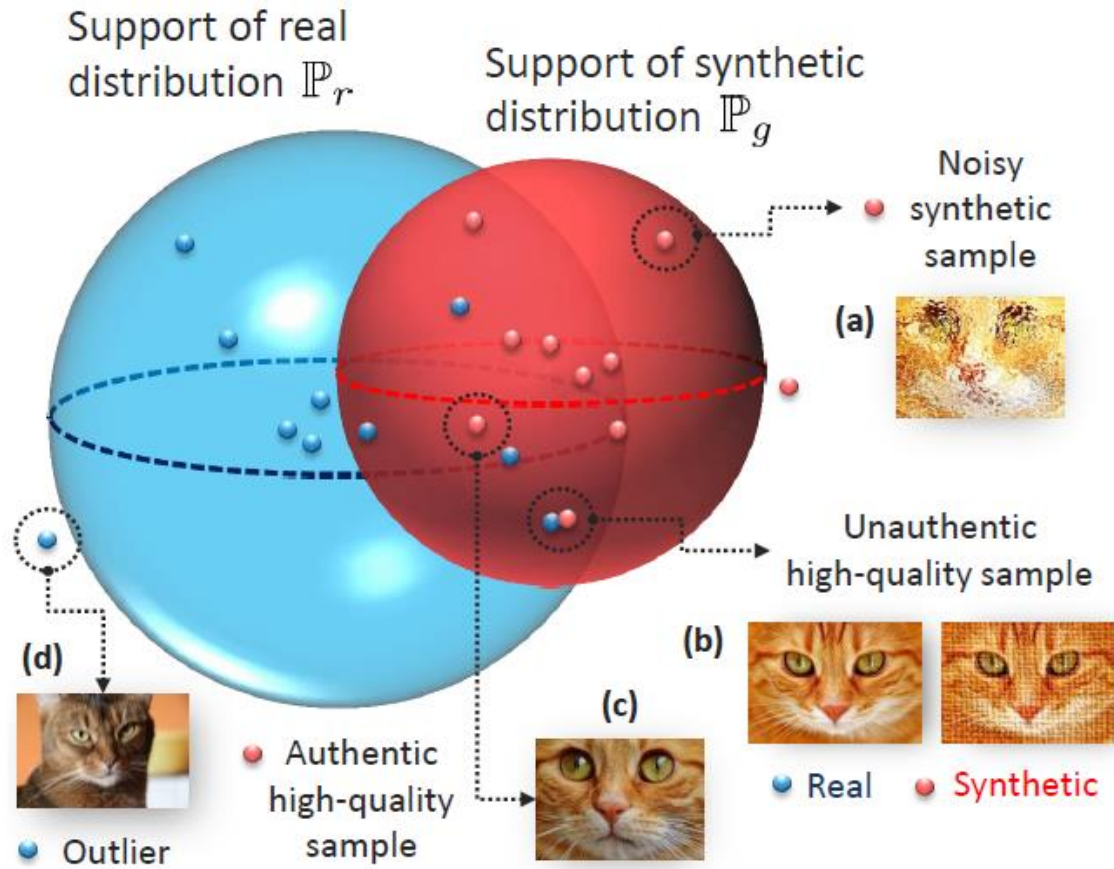


Figure 1. Pictorial depiction for the proposed metrics. The blue and red spheres correspond to the α - and β -supports of real and generative distributions, respectively. Blue and red points correspond to real and synthetic data. (a) Synthetic data falling outside the blue sphere will look unrealistic or noisy. (b) Overfitted models can generate ostensibly high-quality data samples that are “unauthentic” because they are copied from the training data. (c) High-quality data samples should reside inside the blue sphere. (d) Outliers do not count in the β -Recall metric. (Here, $\alpha=\beta=0.9$, α -Precision=8/9, β -Recall = 4/9, and Authenticity = 9/10.)

Project #3: Generative Deep Learning Models for Shapes

Context: Generative deep learning models can be used to generate realistic looking natural images and scenes. In many applications, it is desirable to generate 3D geometric objects (shapes) instead. Many neural network architectures rely on fixed-dimensional vector spaces, but shapes do not naturally yield to a vector space representation. This makes generating shapes using neural networks a challenging problem. Furthermore, using a fixed background grid to represent 3D shapes leads to high-dimensional networks, which are both computationally expensive and difficult to train in a robust manner. One idea to combat this curse of dimensionality is to learn a neural network for shapes using implicit high-frequency functions (Dupont et al. 2022). This leads to resolution independent descriptions that can be used to train for example generative adversarial networks.

This project experiments with generative DL models for shapes. A suitable datasets of shapes should be identified along with, for example predicting rupture risk of brain aneurysms, and generative models developed and validated to extend the dataset for purposes of data augmentation, as one possible example.

Background papers: Dupont et al. 2022

Dataset: Depends on the application, for example AneuX (<https://zenodo.org/records/6678442>)

Project #3: Generative Deep Learning Models for Shapes

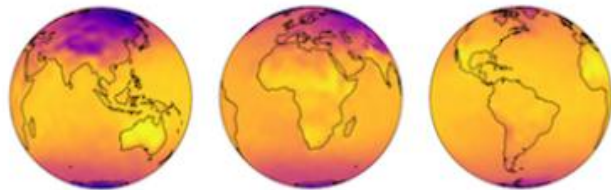
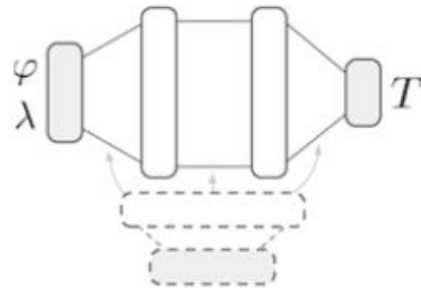
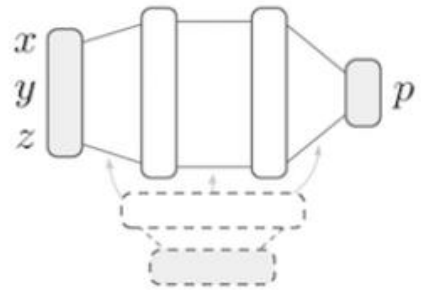
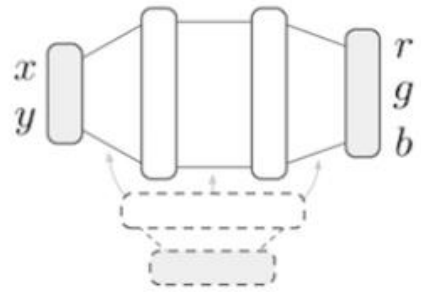


Figure 1: By representing data as continuous functions, we can use the same model to learn distributions of images, 3D shapes and climate data, irrespective of any underlying grid or discretization.

Project #4: Algorithms for Robust Multi-Task Learning in Deep Learning

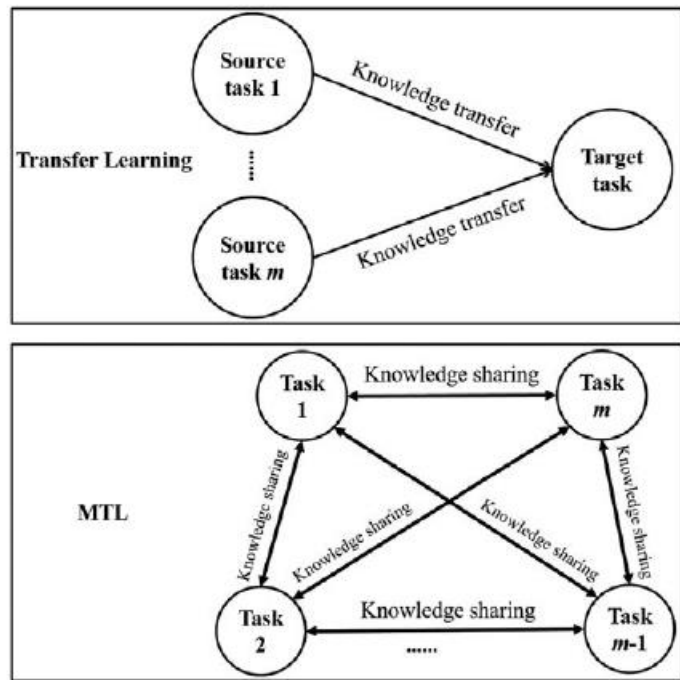
Context: Many machine learning and deep learning tasks require optimizing more than one loss function simultaneously, a challenge known as multi-task learning. This can occur when regularization terms are added to improve convergence for an ill-posed primary task, but it also arises naturally in some applications. Multi-task learning can be difficult because the tasks may have conflicting objectives, leading to gradients of the loss functions that pull in opposite directions. Various algorithmic solutions have been proposed to address multi-task learning problems, with mixed success. Some approaches involve using the concept of Pareto-optimality to identify ensembles of "trade-off" solutions, where improving one loss function would worsen the other.

This project will involve selecting a range of multi-task learning problems, reviewing and proposing different algorithmic solutions, and conducting a comprehensive series of evaluation experiments to identify the most robust approaches. Another focus could be exploring methods for visualizing and explaining the predictions of multi-task learning problems.

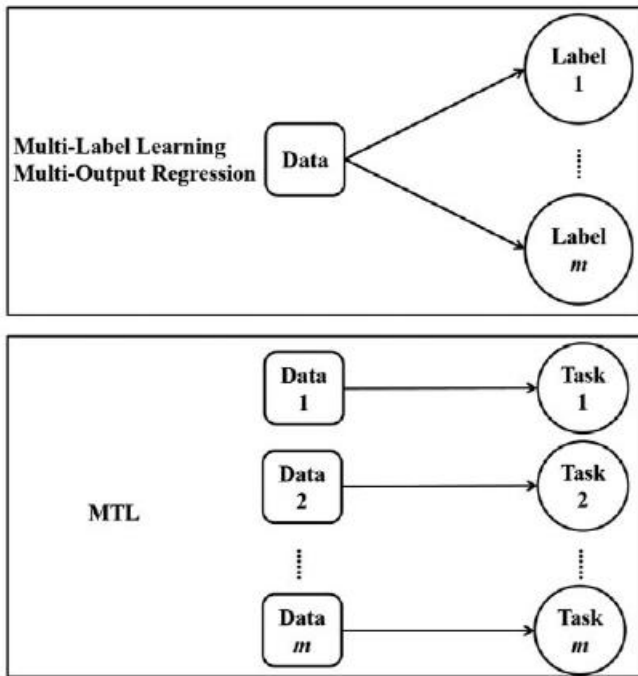
Background papers: Crawshaw 2020, Zhang et al 2022

Dataset: To be selected by the student as part of the project specification

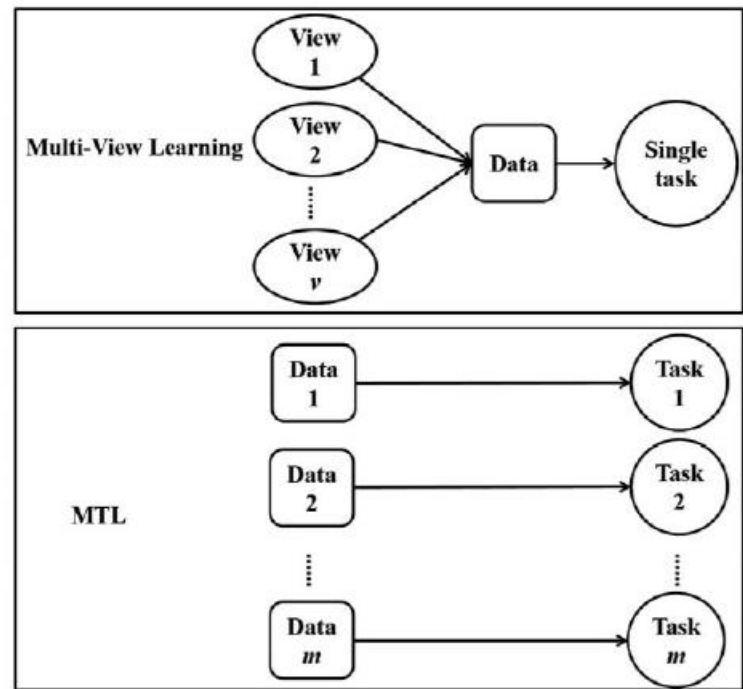
Project #4: Algorithms for Robust Multi-Task Learning in Deep Learning



(a) MTL vs. transfer learning



(b) MTL vs. multi-label learning/multi-output regression



(c) MTL vs. multi-view learning

Fig. 1. Illustrations for differences between MTL and other learning paradigms.

Project #5: Removing Bias from Image Classification Models Using “Unlearning”

Context: Machine learning is used to assist clinical decision making by assessing medical images and providing diagnostic classifications. While as accurate as expert evaluations in many cases, these classifier models can also exhibit unacceptable biases in certain sub-populations, such as women or ethnic minorities. In fact, one can often train a ML model to learn to predict the sex or ethnicity of the subject from just the anatomical image. One possible approach to removing bias from ML models is the so-called “unlearning ” approach – assuming there exists an irrelevant characteristic that doesn’t impact the diagnostic classification, e.g. race, sex, or age of the subject, the ML model is trained to optimise the diagnostic prediction accuracy and simultaneously trained minimise the likelihood that the model can predict the irrelevant characteristic of each the patient. This can lead to more robust, less-biased ML models.

This project explores the use of “unlearning”, for example in the diagnostic classification of chest x-rays where bias in women is known to be present. Using a large dataset of medical images, first a baseline classifier model should be trained and evaluated. Different “unlearning” methods should then be developed and compared against the baseline to see how well such methods can eliminate unwanted biases.

Background papers: Ashraf et al. 2018

Dataset: MedIMeta (<https://zenodo.org/records/7884735>), various others

Project #6: Development of simulation tool for endovascular device deployment

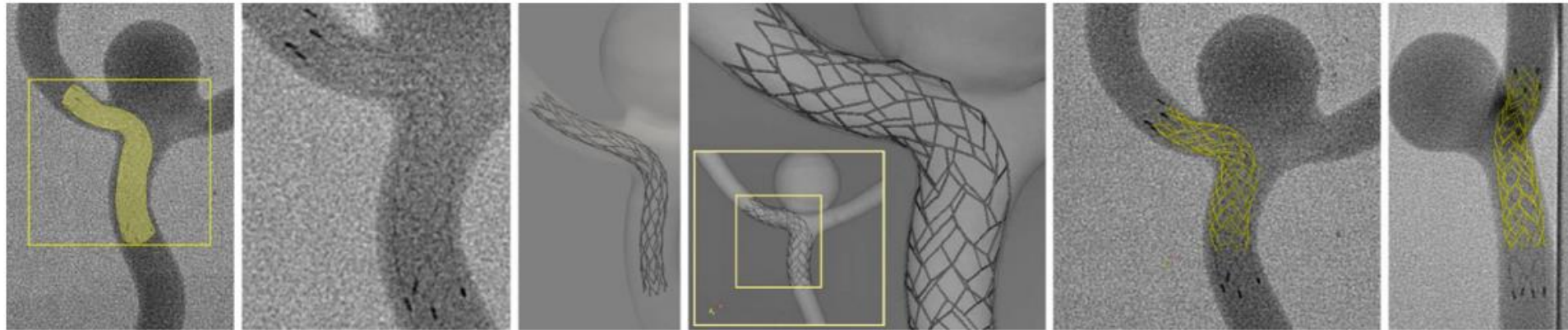
Context: Diseases like coronary artery disease or brain aneurysms can nowadays be treated using minimally invasive endoscopic surgery. An endovascular stent-like device is expanded inside the artery to open an occluded artery or to redirect blood flow away from an aneurysm. Patient-specific computer simulation of endovascular stent deployments can be used to predict treatment success and evaluate the efficacy of novel endovascular devices. However, the stents have very thin metallic struts and accurately modelling their mechanical expansion inside the vessel is computationally challenging. Simplified algorithms for fast virtual deployment are based on mass-spring models and have been shown to give similar predictions in practice.

This project develops a fast virtual deployment software tool that is robust, automated, and uses it to set up and run large-scale in-silico trials for endovascular devices. It uses as a starting point a previously developed open-source tool for virtual device deployment. The project requires good programming skills (Python is sufficient) and basic software engineering skills (agile methodology, collecting requirements, defining solutions, testing).

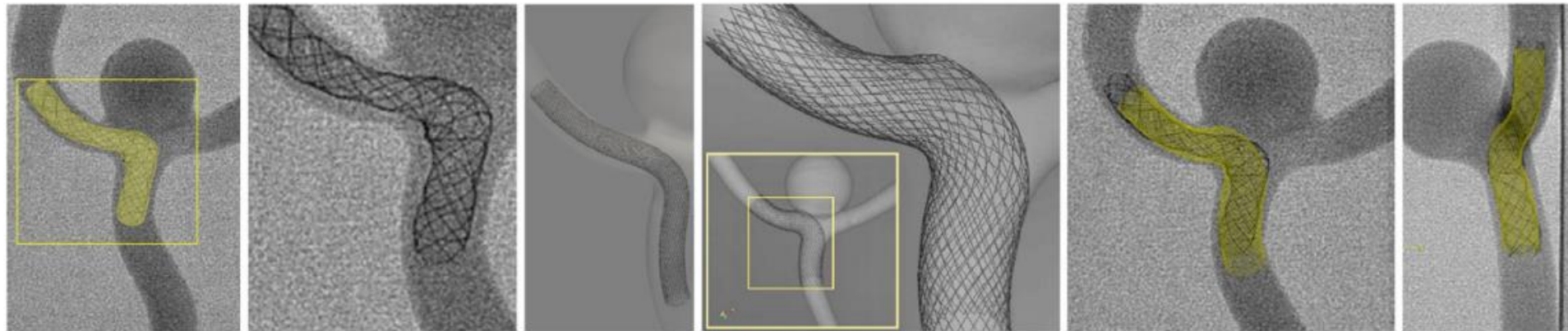
Background papers: Larrabide et al. 2012

Dataset: AneuX (<https://zenodo.org/records/6678442>)

Project #6: Development of simulation tool for endovascular device deployment



(a) *In-vitro* and virtual release (before and after release) for Stent 1. Overlay of virtual stent over *in-vitro* results.



(b) *In-vitro* and virtual release (before and after release) for Stent 2. Overlay of virtual stent over *in-vitro* results.

Fig. 5. Outcome of the experimental and virtual stent release. The images in Fig. 5a correspond to Stent 1 and those in Fig. 5b to Stent 2. Left: image with the highlighted stent and close-up image of the real stent inside the phantom. Center: virtual stent released inside the phantom and close-up for the virtual stent. Right: overlay of the FVS stent on the FD20 images.

Next steps

1. Take a look at the background papers and topics
2. By Monday: Email me your tentative project title
3. Start thinking about aims and objectives, and identify an appropriate dataset
4. Further details on project process to follow

Aims and objectives

Aims:

- High-level goal or the overarching purpose of your research project
- Provides your readers with clarity as to what your study is all about.
- Formulated as a broad statement of the main goal of the research and can range in length from a single sentence to a short paragraph

Aims and objectives

Example of a Research Aim (from discoverphds.com):

“The role of diametrical cup deformation as a factor to unsatisfactory implant performance has not been widely reported.

The aim of this thesis was to gain an understanding of the diametrical deformation behaviour of acetabular cups and shells following impaction into the reamed acetabulum. The influence of a range of factors on deformation was investigated to ascertain if cup and shell deformation may be high enough to potentially contribute to early failure and high wear rates in metal-on-metal implants.”

Aims and objectives

Objectives:

- Detailed steps that your research project will follow.
- Specify how your study will answer the question posed by the aims.
- Formulated as a numbered list where each objective specific begins with an “action verb”, which should be as concrete as possible.

Aims and objectives

Good objectives follow the S.M.A.R.T. rule:

- **S = Specific.** Your objectives should include details of what you want to accomplish and how to achieve them. This includes methods and datasets your project will utilise.
- **M = Measurable.** You should measure your progress and accurately determine whether you've accomplished your objective. A few objectives can be failed if the rest are successfully completed.
- **A = Achievable.** The objective should be achievable within the duration of the project with the skills and resources available to you. Plan ahead if you need access to data or HPC resources.
- **R = Results-based.** The objective should generate concrete outputs that you can put in your report and/or final submitted source code.
- **T = time-limited.** Each objective should have finite duration. Include a Gantt chart in your report.

Aims and objectives

Example of a Research Objectives (from discoverphds.com):

1. *Develop finite element models using explicit dynamics to mimic mallet blows during cup/shell insertion, initially using simplified experimentally validated foam models to represent the acetabulum.*
2. *Investigate the number, velocity and position of impacts needed to insert a cup.*
3. *Determine the relationship between the size of interference between the cup and cavity and deformation for different cup types.*
4. *Investigate the influence of non-uniform cup support and varying the orientation of the component in the cavity on deformation.*
5. *Examine the influence of errors during reaming of the acetabulum which introduce ovality to the cavity.*
6. *Determine the relationship between changes in the geometry of the component and deformation for different cup designs.*
7. *Develop three dimensional pelvis models with non-uniform bone material properties from a range of patients with varying bone quality.*
8. *Use the key parameters that influence deformation, as identified in the foam models to determine the range of deformations that may occur clinically using the anatomic models and if these deformations are clinically significant.*

Ethics

- You will have a session covering ethics and legal issues that underlie student projects
- Some specific considerations for projects that use human or animal-derived datasets:
 - Check that the data collection underwent ethical approval
 - Check whether human data has been anonymised (e.g. no facial pictures of real people)
 - Check whether explicit consent was obtained for use in research
 - Check whether the data use agreement allows use in research
- If you have any doubts, come talk to me.

Literature Searches: Aims

- Choose a limited scope for the project (you can't review everything ever written about machine learning for example)
- Find at least one good review paper that can be used as a basis for background
- Find 2-3 good method papers that can be used as a starting point for implementation

Literature Searches: Techniques

Limit the scope to reduce the number of papers:

1. Come up with relevant keywords to search for (tricky in some cases)
2. Focus on papers published in the last <5 years, exception for clinical studies
3. Review papers can be a useful starting point for finding good method papers

Search engines to use:

1. Scopus (access through UoL network)
2. Google Scholar
3. PubMed (mostly medicine and allied fields)
4. dblp (mostly computer science)
5. MathSciNet (mostly pure maths)

Literature Searches: Techniques

- **Backward citation search:** Start from an interesting paper, see which papers they cite
- **Forward citation search** (Google Scholar): Start from an interesting paper, see which papers cite it
- You can access most journals on University through the library subscription
- **(Experimental):** Ask ChatGPT

Literature Searches: How to evaluate papers efficiently

Review papers:

- Is it a systematic review (meta-analysis) or a narrative review (qualitative interpretation)?
- What are the overall message and findings of the review?
- Where is the field going? What are the open research questions? What are the novel ideas?

Primary research papers:

- Read the abstract, look at figures and tables to understand what was done / is this relevant to me?
- Check for journal quality using SCImago (not guaranteed to imply paper quality)
- Are the methods sufficiently described (neural network architecture, hyperparameters)?
- Is there a reference data set?
- Is there an implementation provided?
- Was the method tested on unseen data? Are the reported results credible?

Computational Resources

Google Colab:

- First choice for prototyping and testing models, free or low-cost plans available

Azure cloud computing:

- Azure Cloud and OpenAI APIs (via Azure) for undergraduate student projects
- Mainly meant for **fine-tuning OpenAI foundation models**, not for heavy computations
- Ask me for access if you feel you could use this in your project

University Advanced Research Cluster (Aire):

- Apply for access at: <https://arc.leeds.ac.uk/platforms/aire/>
- Free to use but busy during semesters
- Interactive sessions limited, need to set up computational environment and launch batch jobs

Planning and Structuring Your Work

- Aims
- Scope
- Identify resources
- Identify tasks
- Draft timeline

Semester 2 – Fortnightly progress meetings

- Prepare to present a 5-minute update on your project
- Huddle: Overview of previous work, current plan, obstacles to progress
- Background: Any material necessary to understand your results
- Results (if any): Keep updating your results as you progress
- Use the progress meetings to prepare materials for your assessor meeting

Example Slide 1 - Huddle

- Progress in the last two weeks
 - Implemented method X on dataset Y
 - Read paper on Z, thinking about how to include in study
 - Started writing Section 2 – Methodology of final report
- Plan for the next two weeks
 - Finalise implementation of X, start planning experiments
 - Investigate new variants for the model
- Issues blocking my progress
 - What is the best way to validate method X? Unsure...

Example Slide 2 - Background

- Present your project in a few sentences
 - Use bulletpoints, not full paragraphs
 - Avoid using too many words. If a slide gets too full, split it into two slides.
 - Define the context of your work first
 - Define any terms or acronyms that are not well-known
 - Present things at a level understandable to a 3rd year undergraduate CS student
- Aims and objectives: Link these to the actual development work
- Define methods if they are non-standard (basic ML/AI methods assumed known)
- Avoid deep technical details (better handled in 1v1 meetings with your supervisor)

Example Slide 3 - Results

- Figure and tables are best for **illustrating results**
- Follow best practice for scientific figures (Mack 2013)
- Each figure should present **one** key finding
- Don't use graphics just to prettify your report
- Avoid using figures from other sources, if possible reproduce them yourself (with citation)
- Tables allow a “deep dive” into your results
- Look at journal papers for ideas how to present results

